

MATH 287
Project #1

For this project, I would like you to work in groups of 2 to perform some analysis of a dataset that is of interest to you. First, you need to identify this dataset, and then use some of the tools we've been learning about during the past few weeks (generally, MLR and model selection) to analyze this data to answer questions of real interest. The data that you use can be from anywhere, and can be about anything – I encourage you to think broadly about the data you might use.

Your group should send me a **brief email by Sunday, October 4th** with information regarding the dataset you are choosing to work on. This email should give me a direct pointer to the data (or attach at least a sample of it) so I can verify that the data is accessible and reasonable (e.g. it should have a quantitative response, and at least 3 [but preferably many more] quantitative predictors). In particular, please make sure it is not simulated data (this has been an issue with some data on Kaggle). To that end, I will also request that you send me some basic summaries of the most important variables (and even some basic visualizations of relationships, as appropriate) to help ensure that the data are authentic.

Assuming I approve your data, you will then have just under two weeks (**due Thursday, Oct. 15**) to do some modelling of the data, and to write a **report to me detailing the analysis and model building process**. This should include diagnostics and transformations, if necessary, as well as model selection. Once a final model is indicated, any interesting inference about this model should be described in the report as well. There is no minimum or maximum page length for the report; make it as long as it needs to be for you to tell me the whole story.

You will have your first WA conference as well for this draft report. Isabel Barsanti will be our WA, and she will be scheduling most of these around Oct 8-9 (+/- a day or two). You should have a decent draft by the time of that meeting, with initial data analyses done, output to frame a discussion, and some process on the model building noted for review. The more you have to show her, the more you can get out of your WA conference!

In addition to a written report, your group will also **present a short description of your data analysis** to the rest of the class during the class meetings on **October 20th and 22nd**. We should be able to have 4 presentations per class, meaning you should aim for a presentation that is around 15 minutes long. That will allow time for Q&A.